

- 6 Bromoethane has a boiling point of 311 K, bromine has a boiling point of 332 K.

Which pair of statements explain this difference?

- 1 The only forces of attraction between bromine molecules are instantaneous dipole-induced dipole attractions.
- 2 The only forces of attraction between bromoethane molecules are permanent dipole-permanent dipole attractions.
- 3 The forces of attraction between bromine molecules are stronger because bromine atoms are more electronegative than carbon and hydrogen atoms.
- 4 The forces of attraction between bromine molecules are stronger because bromine molecules have more electrons.

A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4